MUNICIPAL WETLANDS PERMIT REVIEW ANALYSIS
(MOCK DATASET)

Alyssa Pacheco

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OVERVIEW

Purpose

- Simulate a municipal Conservation Office permit dataset to analyze:
  - Workload distribution
  - Review timelines across wetlands permit types
- The goal is to identify patterns relevant to staffing, process efficiency, and review complexity.

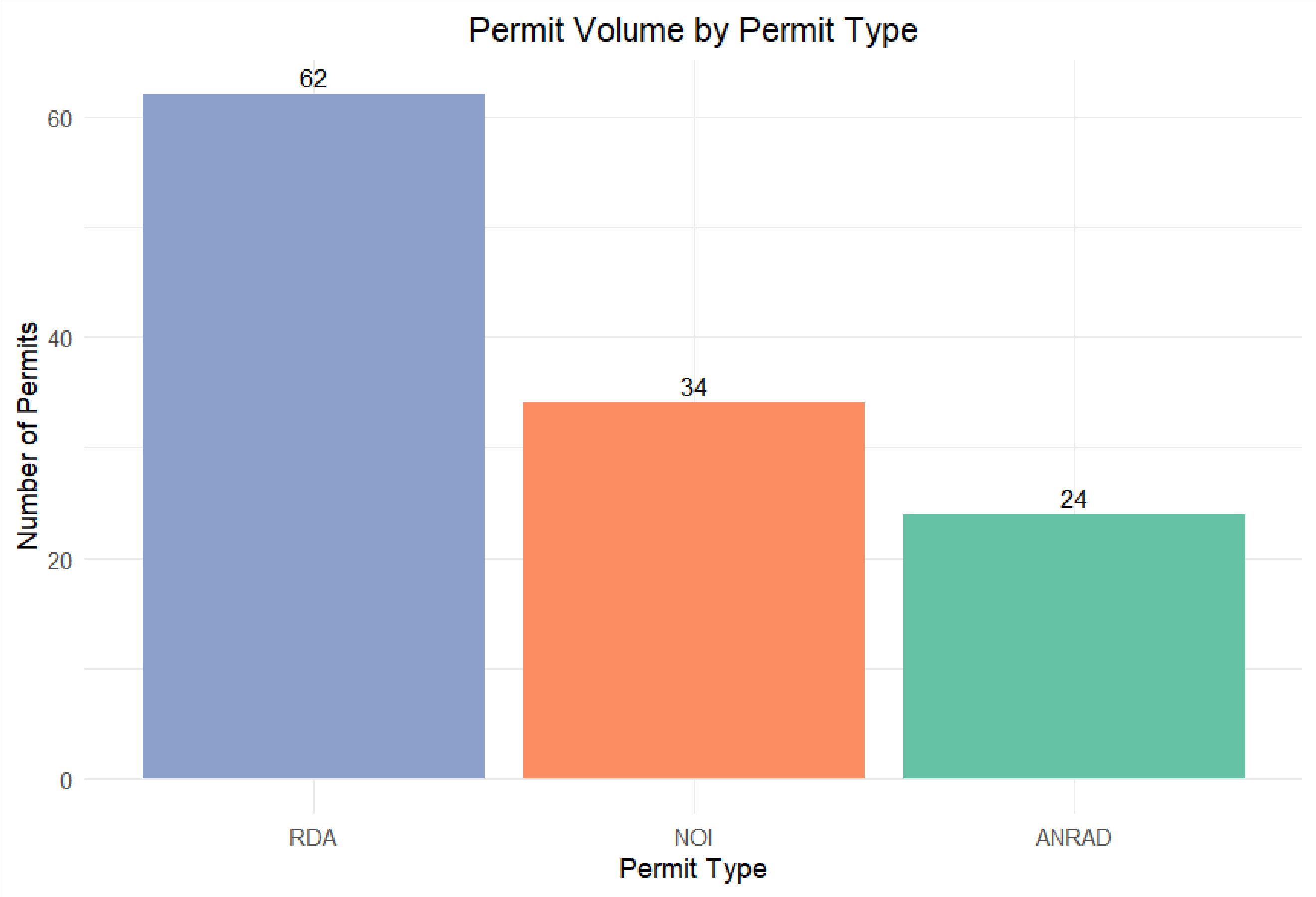
Data Description

- The analysis uses a simulated dataset of 120 wetlands permit records, including:
  - Permit type
  - Wetland resource category
  - Submission and decision dates
  - Permit status
  - Calculated review duration
- Permit distributions and review timelines were modeled to reflect realistic municipal permitting patterns.

Analytical Approach

- Analysis was conducted in R using tidyverse tools to:
  - Summarize permit volume
  - Calculate review timelines
  - Compare patterns across permit types
- Results were evaluated using summary tables and visualizations to assess workload distribution, review variability, and potential process-driven trends.

PERMIT VOLUME & WORKLOAD DISTRIBUTION



What Drives Workload?

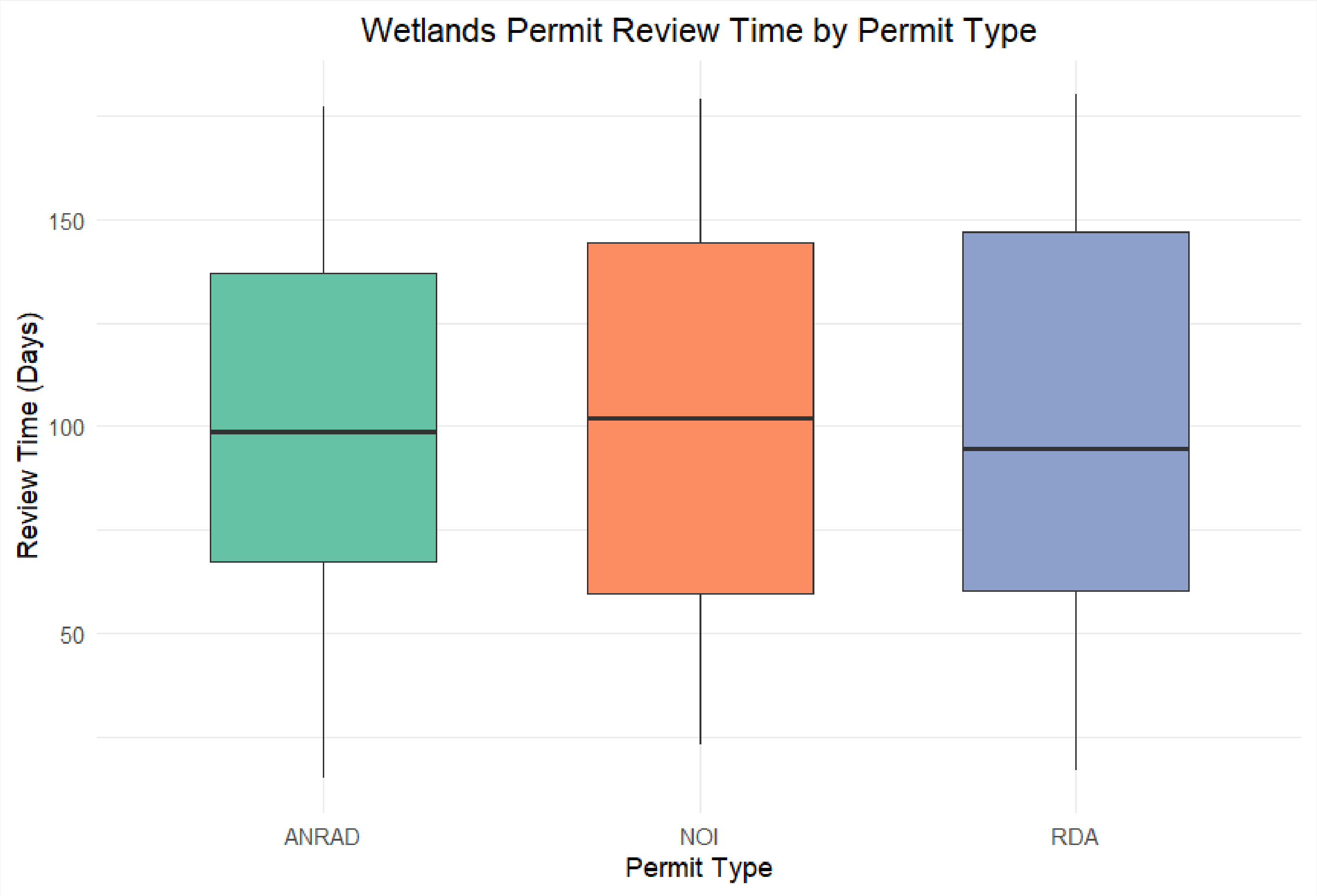
- RDAs account for approximately half of all wetlands permits in the dataset, indicating that Conservation Office workload is primarily driven by determination requests rather than full NOI filings. NOIs represent just over one-quarter of permit activity, while ANRADs comprise a smaller but consistent share of overall workload. This distribution suggests that staff time is concentrated on higher-volume, lower-complexity reviews, with fewer but more involved applications requiring additional effort.

PERMIT VOLUME SUMMARY TABLE

Permit Type	Number of Permits	Percent of Total
RDA	62	51.7%
NOI	34	28.3%
ANRAD	24	20.0%

Table 1. Summarizes permit volume and proportional distribution by permit type.

WETLANDS PERMIT REVIEW TIME BY PERMIT TYPE



How much effort does each permit type require, and where does complexity show up?

- Average review times are similar across permit types, with all categories clustering around approximately 100 days. NOI filings exhibit consistently longer review durations, while RDAs show greater variability, indicating that a subset of determination requests require extended review. Maximum review times approach 180 days across all permit types, suggesting that extended reviews are driven by case-specific factors rather than permit category alone.

REVIEW TIME SUMMARY BY PERMIT TYPE

Permit Type	Permits	Avg Days	Median Days	Min Days	Max Days
NOI	34	101.6	102.0	23	179
RDA	62	100.8	94.5	17	180
ANRAD	24	100.1	98.5	15	177

Table 2. Summary statistics for permit review duration (days) by permit type.